

11 | OVERVIEW



Unabated Warming: Where Does Earth's Excess Heat Go During Surface Temperature Hiatuses?

Abstract

In 2013, many people asked whether global warming had stalled, because global average surface temperatures were rising more slowly than before. Climate scientist Kevin Trenberth explained that warming really means heating, and most of that heat goes into the ocean. His team found that in the previous decade about 30% of the heat had been carried below 700 meters deep, temporarily slowing surface warming but not stopping climate change.

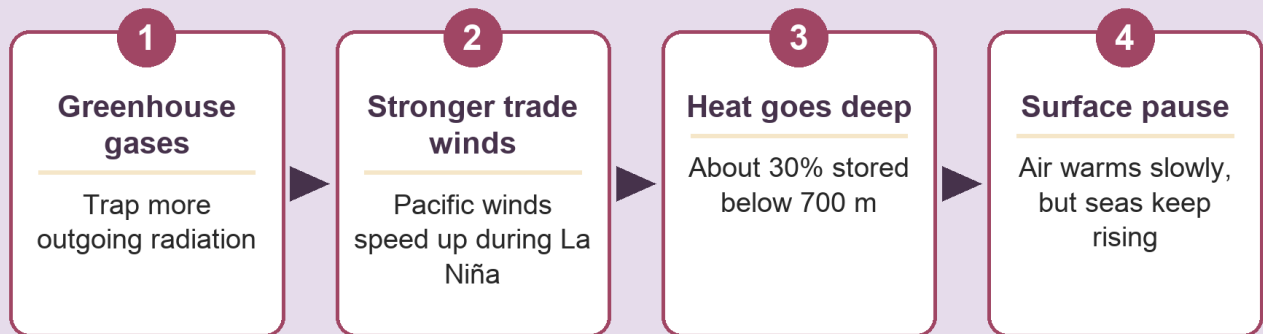
Introduction

Rising carbon dioxide and other greenhouse gases trap more of Earth's outgoing radiation, creating an energy imbalance at the top of the atmosphere. That extra energy shows up in many ways: rising surface temperatures, melting Arctic sea ice, shrinking glaciers, rising seas, and a more intense water cycle. But more than 90% of the energy imbalance goes into the ocean. So when surface temperatures seem to pause, the key question is not whether heating has stopped, but where the heat is going. Trenberth used ocean and temperature records to show that a slowdown at the surface does not mean the planet has stopped warming.

How Do Scientists Study This?

Trenberth and colleagues analyzed ocean heat content at different depths. Many earlier analyses only looked at the top 700 meters of the ocean, so the team extended their analysis deeper. They then linked changes in where heat was stored to changes in winds, especially the Pacific trade winds, and to natural climate patterns such as El Niño and La Niña. They also considered other natural factors that affect Earth's energy balance, including volcanic eruptions and the Sun's 11-year sunspot cycle. Finally, they examined the NOAA global temperature record to compare short flat periods with the long-term warming trend.

Where the heat went during the hiatus



Look deeper

Scientists measured ocean heat below 700 meters, where most earlier analyses stopped.

Find the mechanism

Stronger Pacific trade winds changed currents, carrying heat down into the deep ocean.

Check the long view

The global temperature record was examined for decades, not just a few years.

11 | FINDINGS

What Did They Discover?

In the decade before 2013, about 30% of the ocean's added heat was stored below 700 meters. This was new, caused by stronger Pacific trade winds linked to La Niña conditions. Some of this deep heat returns during the next El Niño, producing a mini global warming at the surface, but much stays in the deep ocean. That means less short-term surface warming but greater long-term warming and faster sea-level rise. The Sun's quiet phase from 2005 to 2010 also reduced the energy imbalance by about 10 to 15%. Over the long term, warming since the 1970s has averaged about 0.16°C per decade, with flat periods from 1977 to 1986, 1987 to 1996, and 2001 to 2012, each followed by a big jump.

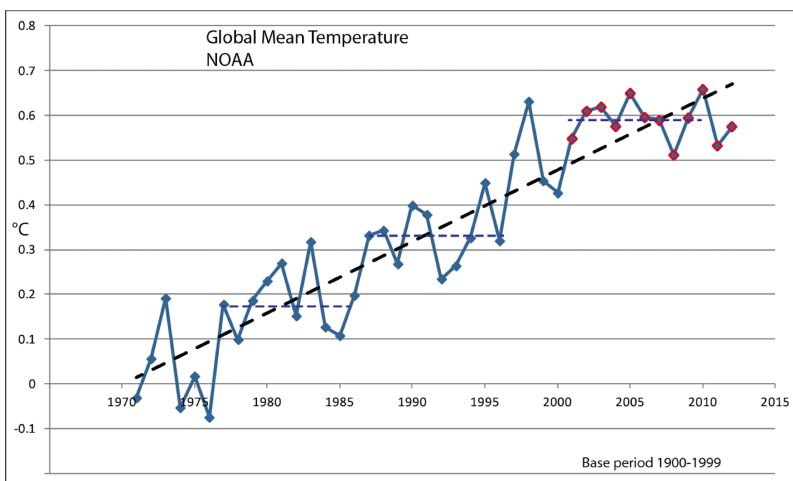
Warming by the numbers

90%+ of the energy imbalance goes into the ocean

30% of recent ocean heat stored below 700 m

0.16°C average warming per decade since the 1970s

Global sea level has risen at more than 30 cm per century since satellite measurements began in 1992, a steady sign that warming continues.



FROM THE ARTICLE

Global mean temperature relative to 1900 to 1999, with the long-term trend (dashed). Source: NOAA, as shown in the article.

11 | TAKEAWAYS

Takeaways

Heat hides in the ocean

More than 90% of trapped energy goes into the ocean, not the air.

Pauses are temporary

Flat periods in surface warming occur, but each is followed by a jump.

Look at the big picture

Sea-level rise and ocean heat show warming continues unabated.

Key Terms

Energy imbalance

The difference between energy Earth absorbs from the Sun and energy it sends back to space. A positive imbalance causes heating.

Hiatus

A period when global surface temperature rises more slowly or levels off, even though the planet keeps gaining heat.

Ocean heat content

The total amount of heat stored in the ocean, measured at different depths.

Trade winds

Steady winds that blow from east to west near the equator. Changes in their strength affect ocean currents and heat storage.

El Niño and La Niña

Opposite phases of a natural Pacific climate pattern that shift heat between the ocean and atmosphere.

11 | ACTIVITY

Activity

Stairs, Not a Ramp

Students analyze temperature data to see how short flat periods fit within a long-term warming trend.

Materials: Printed NOAA global temperature graph from the lesson; Rulers; Colored pencils

1. On the graph, draw a straight line showing the overall warming trend since 1970.
2. Shade the three hiatus periods: 1977 to 1986, 1987 to 1996, and 2001 to 2012.
3. Describe what happens to temperature at the end of each flat period.
4. Explain in your own words where the heat went during the most recent hiatus.
5. Debate: why is sea-level rise a better indicator of continued warming than a few years of surface temperature?

References

Trenberth, K. (2013). Global warming is here to stay, whichever way you look at it. The Conversation.
<https://doi.org/10.64628/AB.hf49n9mx5>